European Network of
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for Electricity

entsoe

File Header Guidelines

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1. Introduction

The document is providing guidance on how to apply IEC 61970-552 which is defining file headers for complete instance files and difference instance files. These guidelines are applied when producing the Test Configurations applicable for the ENTSO-E Common Grid Model Exchange Standard (CGMES) Conformity Assessment Framework hereafter referred as "the Framework".

Versioning of the document is following the rules specified in the Chapter 5.1 of the CGMES.

2. Exchange scenarios

The IEC 61970-552 gives as an example the following figure.

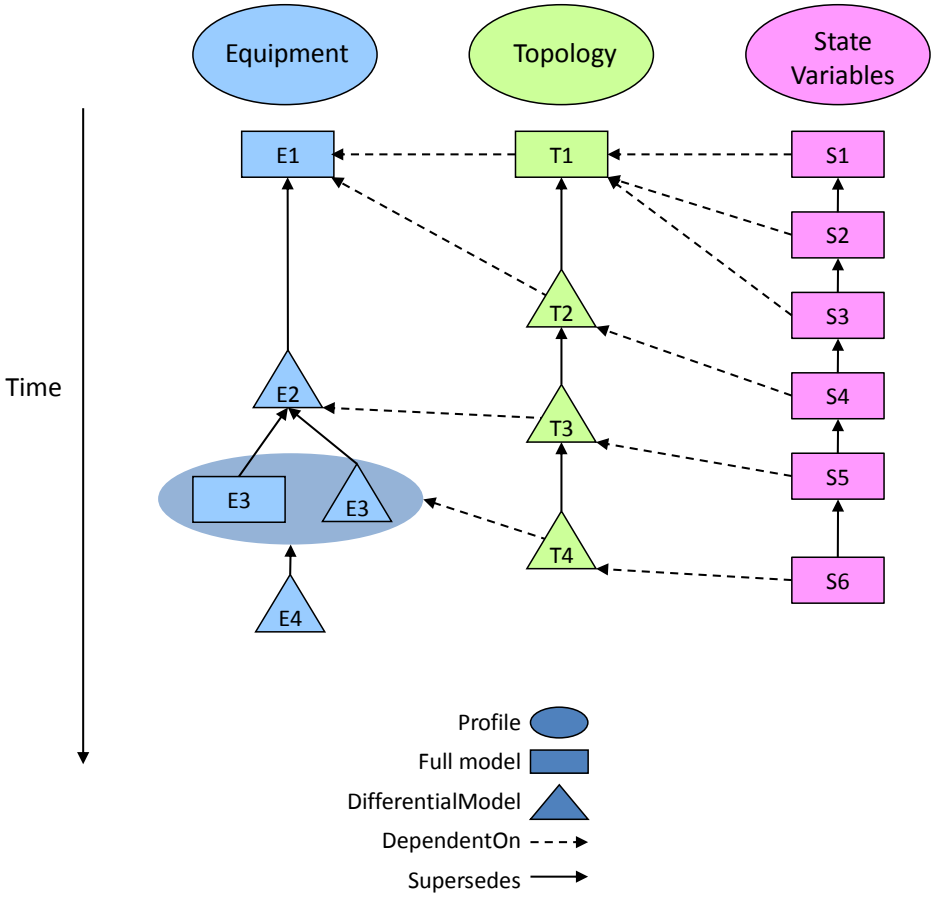


Figure 1 – Example of model dependencies

Figure 1 is used in order to explain what information should be provided in different file headers.

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3. Examples

3.1. Example 1: File header of full model

This example is for full model E1 from Figure 1.

Syntax	Comment/Explanation
<?xml version="1.0"?> <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:cim="http://iec.ch/TC57/2013/CIM-schema-cim16#" xmlns:md="http://iec.ch/TC57/61970-552/ModelDescription/1#" xmlns:entsoe=http://entsoe.eu/CIM/SchemaExtension/3/1#>	
<md:FullModel rdf:about="urn:uuid:bcb6877a-e948-11e3-89cf-82687f4fc15c">	This is the ID of the model included in the instance file. Please refer to Chapter 4.3 in the CGMES. The ID to which other models (full or difference) refer in case of depend on or supersede. Please note that the "urn:uuid:" is used only for rdf:about or rdf:resource in the file header, i.e. it is not applied for other elements in the instance file.
<md:Model.scenarioTime> 2030-01-15T17:00:00.000</md:Model.scenarioTime>	This is the date & time that this model represents, i.e. for which the model is valid.
<md:Model.created>2014-05-15T17:48:31.474</md:Model.created>	The date & time when the model was created. It is the time of the serialization.
<md:Model.description>CGMES Conformity Assessment: This is guidelines on the file header.</md:Model.description>	A description of the model and explanation of the purpose.
<md:Model.version>1</md:Model.version>	Version of the model. If the instance file is imported and exported with no change the version number is the kept same. The version changes only if the content of the file changes. It is the same logic as for the header id (see CGMES [R.4.3.4.1.]). The version is the human readable id. The version is integer number.
<md:Model.profile>http://entsoe.eu/CIM/EquipmentCore/3/1</md:Model.profile>	One to many URN/URI describing the profiles that governs this model. It uniquely identifies the profiles and its version. In this case EQ instance file contains one profile.
<md:Model.profile>http://entsoe.eu/CIM/EquipmentOperation/3/1</md:Model.profile> <md:Model.profile>http://entsoe.eu/CIM/EquipmentShortCircuit/3/1</md:Model.profile>	Additional profile URN/URI are included if classes and attributes that belong to Operation and Short Circuit are used and the instance file is intended for such use.
<md:Model.modelingAuthoritySet>http://elia.be/Planning/CGMES/2.4.14</md:Model.modelingAuthoritySet> </md:FullModel>	A URN/URI referring to the organisation role / model authority set reference. The organization role is the source of the model. It is the same for all profiles part of a model exchange. This concept might be modified in the future when applying the concept of "frames" which is under discussion.

3.2. Example 2: File header of full model that is depending on another model

This example is for full model T1 from Figure 1.

Syntax	Comment/Explanation
<?xml version="1.0"?> <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:cim="http://iec.ch/TC57/2013/CIM-schema-cim16#" xmlns:md="http://iec.ch/TC57/61970-552/ModelDescription/1#" xmlns:entsoe=http://entsoe.eu/CIM/SchemaExtension/3/1#>	
<md:FullModel rdf:about="urn:uuid:d63e4784-e94b-11e3-89cf-82687f4fc15c">	This is the ID of the model included in the instance file. Please refer to Chapter 4.3 in the CGMES. The ID to which other models (full or difference) refer in case of depend on or supersede. Please note that the "urn:uuid:" is used only for rdf:about or rdf:resource in the file header, i.e. it is not applied for other elements in the instance file.
<md:Model.scenarioTime> 2030-01-15T17:00:00.000</md:Model.scenarioTime>	This is the date & time that this model represents, i.e. for which the model is valid.
<md:Model.created>2014-05-15T17:48:31.474</md:Model.created>	The date & time when the model was created. It is the time of the serialization.
<md:Model.description>CGMES Conformity Assessment: This is guidelines on the file header.</md:Model.description>	A description of the model and explanation of the purpose.
<md:Model.version>1</md:Model.version>	Version of the model. If the instance file is imported and exported with no change the version number is the kept same. The version changes only if the content of the file changes. It is the same logic as for the header id (see CGMES [R.4.3.4.1.]). The version is the human readable id. The version is integer number.
<md:Model.profile>http://entsoe.eu/CIM/Topology/4/1</md:Model.profile>	One to many URN/URI describing the profiles that governs this model. It uniquely identifies the profiles and its version. In this case it is the profile URI of the TP profile.
<md:Model.DependentOn rdf:resource="urn:uuid:bcb6877a-e948-11e3-89cf-82687f4fc15c" />	A reference to the model documents that the model described by this document depends on, e.g. in this case the T1 model depends on E1 model. In general there can be 0 or many Model.DependentOn depending on the profile and the content of the instance file.
<md:Model.modelingAuthoritySet>http://elia.be/Planning/CGMES/2.4.14</md:Model.modelingAuthoritySet> </md:FullModel>	A URN/URI referring to the organisation role / model authority set reference. The organization role is the source of the model. It is the same for all profiles part of a model exchange. This concept might be modified in the future when applying the concept of "frames" which is under discussion.

3.3. Example 3: File header of full model that is depending on a model and supersedes another model

This example is for full model S2 from Figure 1.

Syntax	Comment/Explanation
<?xml version="1.0"?> <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:cim="http://iec.ch/TC57/2013/CIM-schema-cim16#" xmlns:md="http://iec.ch/TC57/61970-552/ModelDescription/1#" xmlns:entsoe=http://entsoe.eu/CIM/SchemaExtension/3/1#>	
<md:FullModel rdf:about="urn:uuid:aad4c9cc-e94d-11e3-89cf-82687f4fc15c">	This is the ID of the model included in the instance file. Please refer to Chapter 4.3 in the CGMES. The ID to which other models (full or difference) refer in case of depend on or supersede. Please note that the "urn:uuid:" is used only for rdf:about or rdf:resource in the file header, i.e. it is not applied for other elements in the instance file.
<md:Model.scenarioTime> 2030-01-15T17:00:00.000</md:Model.scenarioTime>	This is the date & time that this model represents, i.e. for which the model is valid.
<md:Model.created>2014-05-15T17:48:31.474</md:Model.created>	The date & time when the model was created. It is the time of the serialization.
<md:Model.description>CGMES Conformity Assessment: This is guidelines on the file header.</md:Model.description>	A description of the model and explanation of the purpose.
<md:Model.version>2</md:Model.version>	Version of the model. If the instance file is imported and exported with no change the version number is the kept same. The version changes only if the content of the file changes. It is the same logic as for the header id (see CGMES [R.4.3.4.1.]). The version is the human readable id. The version is integer number.
<md:Model.profile>http://entsoe.eu/CIM/StateVariables/4/1</md:Model.profile>	One to many URN/URI describing the profiles that governs this model. It uniquely identifies the profiles and its version. In this case it is the profile URI of the SV profile.
<md:Model.DependentOn rdf:resource="urn:uuid:d63e4784-e94b-11e3-89cf-82687f4fc15c" />	A reference to the model documents that the model described by this document depends on, e.g. in this case the S2 model depends on T1 model. In general there can be 0 or many Model.DependentOn depending on the profile and the content of the instance file.
<md:Model.Supersedes rdf:resource="urn:uuid:b1f4d506-e94f-11e3-89cf-82687f4fc15c"/>	When a model is updated the resulting model supersedes the models that were used as basis for the update. Hence this is a reference to the CIMXML documents which are superseded by this model. A model (or instance file) can supersede 1 or more models, e.g. a difference model or a full model supersede multiple models (difference or full). In this case more than one Model.Supersedes are included in the header. The referenced document(s) is (are) identified by the URN/MRID/UUID in the FullModel rdf:about attribute when full model(s) is (are) referenced and by the URN/MRID/UUID in the DifferenceModel rdf:about attribute when difference model(s) is (are) referenced. In this case the model S2 supersedes model S1 which has <md:FullModel rdf:about="urn:uuid:b1f4d506-e94f-11e3-89cf-82687f4fc15c">
<md:Model.modelingAuthoritySet>http://elia.be/Planning/CGMES/2.4.14</md:Model.modelingAuthoritySet> </md:FullModel>	A URN/URI referring to the organisation role / model authority set reference. The organization role is the source of the model. It is the same for all profiles part of a model exchange. This concept might be modified in the future when applying the concept of "frames" which is under discussion.

3.4. Example 4: File header of difference model that is depending on a full model and supersedes another full model

This example is for difference model T2 from Figure 1.

Syntax	Comment/Explanation
<?xml version="1.0"?> <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:cim="http://iec.ch/TC57/2013/CIM-schema-cim16#" xmlns:dm="http://iec.ch/TC57/61970-552/DifferenceModel/1#" xmlns:md="http://iec.ch/TC57/61970-552/ModelDescription/1#" xmlns:entsoe=http://entsoe.eu/CIM/SchemaExtension/3/1#>	Please pay attention that for difference models xmlns:dm=http://iec.ch/TC57/61970-552/DifferenceModel/1# is included as well
<dm:DifferenceModel rdf:about="urn:uuid:7ca72efa-e952-11e3-89cf-82687f4fc15c">	This is the ID of the model included in the instance file. Please refer to Chapter 4.3 in the CGMES. The ID to which other models (full or difference) refer in case of depend on or supersede. Please note that the "urn:uuid:" is used only for rdf:about or rdf:resource in the file header, i.e. it is not applied for other elements in the instance file.
<md:Model.scenarioTime> 2030-01-15T17:00:00.000</md:Model.scenarioTime>	This is the date & time that this model represents, i.e. for which the model is valid.
<md:Model.created>2014-05-15T17:48:31.474</md:Model.created>	The date & time when the model was created. It is the time of the serialization.
<md:Model.description>CGMES Conformity Assessment: This is guidelines on the file header.</md:Model.description>	A description of the model and explanation of the purpose.
<md:Model.version>2</md:Model.version>	Version of the model. If the instance file is imported and exported with no change the version number is the kept same. The version changes only if the content of the file changes. It is the same logic as for the header id (see CGMES [R.4.3.4.1.]). The version is the human readable id. The version is integer number.
<md:Model.profile>http://entsoe.eu/CIM/Topology/4/1</md:Model.profile>	One to many URN/URI describing the profiles that governs this model. It uniquely identifies the profiles and its version. In this case it is the profile URI of the TP profile.
<md:Model.DependentOn rdf:resource="urn:uuid:bcb6877a-e948-11e3-89cf-82687f4fc15c" />	A reference to the model documents that the model described by this document depends on, e.g. in this case the T2 model depends on E1 model. In general there can be 0 or many Model.DependentOn depending on the profile and the content of the instance file.
<md:Model.Supersedes rdf:resource="urn:uuid:d63e4784-e94b-11e3-89cf-82687f4fc15c"/>	When a model is updated the resulting model supersedes the models that were used as basis for the update. Hence this is a reference to the CIMXML documents which are superseded by this model. A model (or instance file) can supersede 1 or more models, e.g. a difference model or a full model supersede multiple models (difference or full). In this case more than one Model.Supersedes are included in the header. The referenced document(s) is (are) identified by the URN/MRID/UUID in the FullModel rdf:about attribute when full model(s) is (are) referenced and by the URN/MRID/UUID in the DifferenceModel rdf:about attribute when difference model(s) is (are) referenced. In this case the model T2 supersedes model T1, i.e. this difference model is applied on the model which has the id mentioned in this Model.Supersedes.
<md:Model.modelingAuthoritySet>http://elia.be/Planning/CGMES/2.4.14</md:Model.modelingAuthoritySet>	A URN/URI referring to the organisation role / model authority set reference. The organization role is the source of the model. It is the same for all profiles part of a model exchange. This concept might be modified in the future when applying the concept of "frames" which is under discussion.
Here the place for the content of the difference file	This follows the syntax as defined in the 61970-452.
</dm:DifferenceModel>	

3.5. Example 5: File header of difference model that is depending on a difference model and supersedes another difference model

This example is for difference model T3 from Figure 1.

Syntax	Comment/Explanation
<?xml version="1.0"?> <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:cim="http://iec.ch/TC57/2013/CIM-schema-cim16#" xmlns:dm="http://iec.ch/TC57/61970-552/DifferenceModel/1#" xmlns:md="http://iec.ch/TC57/61970-552/ModelDescription/1#" xmlns:entsoe=http://entsoe.eu/CIM/SchemaExtension/3/1#>	Please pay attention that for difference models xmlns:dm=http://iec.ch/TC57/61970-552/DifferenceModel/1# is included as well
<dm:DifferenceModel rdf:about="urn:uuid:80323db8-ea73-11e3-9df6-82687f4fc15c">	This is the ID of the model included in the instance file. Please refer to Chapter 4.3 in the CGMES. The ID to which other models (full or difference) refer in case of depend on or supersede. Please note that the "urn:uuid:" is used only for rdf:about or rdf:resource in the file header, i.e. it is not applied for other elements in the instance file.
<md:Model.scenarioTime> 2030-01-15T17:00:00.000</md:Model.scenarioTime>	This is the date & time that this model represents, i.e. for which the model is valid.
<md:Model.created>2014-05-15T17:48:31.474</md:Model.created>	The date & time when the model was created. It is the time of the serialization.
<md:Model.description>CGMES Conformity Assessment: This is guidelines on the file header.</md:Model.description>	A description of the model and explanation of the purpose.
<md:Model.version>3</md:Model.version>	Version of the model. If the instance file is imported and exported with no change the version number is the kept same. The version changes only if the content of the file changes. It is the same logic as for the header id (see CGMES [R.4.3.4.1.]). The version is the human readable id. The version is integer number.
<md:Model.profile>http://entsoe.eu/CIM/Topology/4/1</md:Model.profile>	One to many URN/URI describing the profiles that governs this model. It uniquely identifies the profiles and its version. In this case it is the profile URI of the TP profile.
<md:Model.DependentOn rdf:resource="urn:uuid:21a419b4-ea74-11e3-9df6-82687f4fc15c" />	A reference to the model documents that the model described by this document depends on, e.g. in this case the T3 model depends on difference model E2 which has <dm:DifferenceModel rdf:about="urn:uuid:21a419b4-ea74-11e3-9df6-82687f4fc15c">. In general there can be 0 or many Model.DependentOn depending on the profile and the content of the instance file.
<md:Model.Supersedes rdf:resource="urn:uuid:7ca72efa-e952-11e3-89cf-82687f4fc15c"/>	When a model is updated the resulting model supersedes the models that were used as basis for the update. Hence this is a reference to the CIMXML documents which are superseded by this model. A model (or instance file) can supersede 1 or more models, e.g. a difference model or a full model supersede multiple models (difference or full). In this case more than one Model.Supersedes are included in the header. The referenced document(s) is (are) identified by the URN/MRID/UUID in the FullModel rdf:about attribute when full model(s) is (are) referenced and by the URN/MRID/UUID in the DifferenceModel rdf:about attribute when difference model(s) is (are) referenced. In this case the model T3 supersedes the difference model T2, i.e. this difference model is applied on the model which has the id mentioned in this Model.Supersedes.
<md:Model.modelingAuthoritySet>http://elia.be/Planning/CGMES/2.4.14</md:Model.modelingAuthoritySet>	A URN/URI referring to the organisation role / model authority set reference. The organization role is the source of the model. It is the same for all profiles part of a model exchange. This concept might be modified in the future when applying the concept of "frames" which is under discussion.
Here the place for the content of the difference file	This follows the syntax as defined in the 61970-452.
</dm:DifferenceModel>	